

BANARAS HINDU UNIVERSITY
B.Sc. (Hons.) Semester I Examination, 2016-17

Paper No. STB-101: Statistical Methods and Probability

Time: 3 Hours

Maximum Marks: 70

*Instructions: Answer **five** questions including Question 1 which is compulsory. Symbols have their usual meanings.*

Question 1.

- (a) What are the different types of scales of measurements? Explain each with a suitable example.
- (b) Describe primary and secondary data. State their relative merits and defects.
- (c) Explain different parts of a table.

Question 2.

- (a) Explain histogram, frequency polygon and ogives with examples and diagrams.
- (b) Two sets of data consisting of n_1 and n_2 values have medians M_1 , M_2 and means $\bar{X}_1$ and $\bar{X}_2$. Show that the combined median lies between M_1 and M_2 , and the combined mean also lies between $\bar{X}_1$ and $\bar{X}_2$.

Question 3.

- (a) For discrete data, prove that the mean deviation is least when it is measured from the median.
- (b) Prove that the mean deviation about the mean $\bar{x}$ for a frequency distribution $(x_i, f_i; i = 1, 2, \dots, n)$ is $\frac{2}{N} \sum_{x_i < \bar{x}} f_i (\bar{x} - x_i)$, where $N = \sum_{i=1}^n f_i$.

Question 4.

- (a) Define raw and central moments. Define coefficients $\beta_1, \beta_2, \gamma_1$ and γ_2 . Prove that $\beta_2 \geq 1$.
- (b) When is a frequency distribution said to be a symmetrical distribution? Explain skewness and its different measures.

Question 5.

- (a) Give the axiomatic definition of probability. Using its axioms, prove that:
- (i) $P(A \cup B) = P(A) + P(B) - P(A \cap B)$
 - (ii) If $B \subset A$, then $P(B) \leq P(A)$
- (b) The chances of winning of two race horses are a and b respectively. What is the probability that at least one will win when the horses are running:
- (i) in different races,
 - (ii) in the same race.

Question 6.

- (a) A man alternately tosses a coin and throws a die, beginning with the coin. What is the probability that he will get a head before he gets a 5 or 6 on the die?
- (b) An urn contains 6 white and 4 black balls, and a second urn contains 7 white and 2 black balls. A ball is randomly transferred from the first urn to the second urn, and then a ball is randomly transferred from the second urn back to the first urn. Now a ball is drawn from the first urn. What is the probability that it is white?
- (c) The chances of A, B, and C becoming managers of a certain company are $4 : 2 : 3$. The probability that a bonus scheme will be introduced if A, B, or C becomes manager is 0.3, 0.5, and 0.8 respectively. If the bonus has been introduced, what is the probability that A was appointed as the manager?

Question 7.

- (a) Define a random variable. A random variable X has the following probability mass function:

x	0	1	2	3
$f(x)$	0.1	K	$2K$	0.3

- (i) Find the value of K , $E(X)$ and $V(X)$.
 - (ii) Find the distribution function of X and draw its graph.
- (b) Define joint probability mass function (p.m.f.) of two discrete random variables. If the joint p.m.f. of X_1 and X_2 is:

$$f(x_1, x_2) = \frac{x_1 + x_2}{21}, \quad x_1 = 1, 2, 3; \quad x_2 = 1, 2$$

and 0 otherwise. Find the marginal p.m.f. of X_1 and X_2 , and the conditional p.m.f. of X_1 given $X_2 = 2$.

Question 8.

- (a) Two random variables X and Y have the joint probability density function:

$$f(x, y) = \frac{1}{8}(6 - x - y), \quad 0 < x < 2, \quad 2 < y < 4$$

and 0 otherwise. Find:

(i) $P[X < 1 \text{ and } Y < 3]$

(ii) $P[X + Y < 3]$

- (b) If (X, Y) has the joint p.d.f.:

$$f(x, y) = kxy, \quad 0 < x < y < 1$$

and 0 otherwise. Find:

(i) the value of k ,

(ii) the marginal and conditional p.d.f.s.